

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
 DURATION: 3 HOURS

SUMMER SEMESTER, 2024-2025
 FULL MARKS: 150

CSE 4803: Graph Theory

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Suppose you are working in a factory where there is a network of 4 assemblers, labelled a_1 through a_4 , connected by conveyor belts. You are given the following adjacency matrix representing the network:

$$\begin{matrix} & a_1 & a_2 & a_3 & a_4 \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{matrix} & \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \end{matrix}$$

- a) Analyze whether this assembler network is disconnected, weakly connected, or strongly connected. 5
(CO2)
(PO2)
- b) Find the number of distinct sequences of conveyor belts of length 4 between assemblers a_4 and a_1 . 8
(CO1)
(PO1)
- 6 c) The factory owner wants to know the total number of distinct ways an item may travel from a starting assembler and come back to the starting assembler, after being processed by all other assemblers while travelling through all the conveyor belts exactly once. 12
(CO3)
(PO3)
- Recommend a solution for this problem, showing the detailed steps of your method.

2. Suppose you are designing a smart irrigation system using a controller with 5 binary inputs, labelled A through E. These inputs are coming from specialized sensors that can detect the following:

- A: Soil is dry
- B: Temperature is high
- C: Humidity is low
- D: Rain is not detected
- E: Sunlight is high

You have set the irrigation system to activate based on the following Boolean function:

$$F(A, B, C, D, E) = AB'C'D'E + AB'C'DE + AB'CD'E + AB'CDE + ABC'D'E + ABC'DE + ABCD'E + ABCDE$$

After deploying the system, you notice that there is a redundancy since not all of the sensors are actually needed. Thus, you decide to reduce your Boolean function.

- a) Formulate the scenario as a graph theory problem and explain the significance of the vertices and edges for such a problem. 6
(CO2)
(PO2)

b) Refer to the graph formulated in Question 2.a before reducing the Boolean function. Justify whether a perfect matching exists for that graph.

c) Based on the problem formulation in Question 2.a, recommend the minimum number of inputs for the Boolean function to work as intended.

F, AF

3. a) Prove by construction that the regular bipartite graph of six vertices, $K_{3,3}$, is nonplanar.

b) Show whether the two graphs shown in Figure 1 are duals of the same original graph.

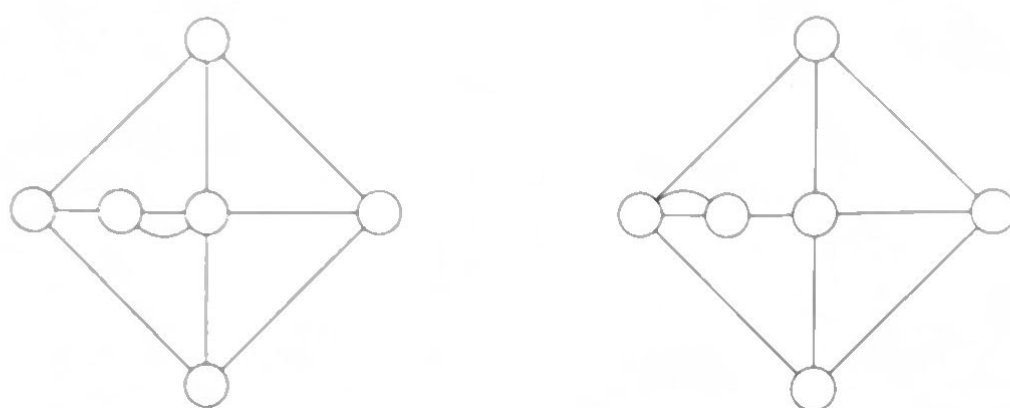


Figure 1: Sample dual graphs for Question 3.b

c) If every region of a simple planar graph of n vertices and e edges embedded in a plane is bounded by the same number of edges, k , then establish a relationship among them expressing e as a function of n and k .

$$e \leq 2k(n-2)$$

4. a) Figure 2 shows a graph representing five students, labelled s_1 through s_5 , seated in a circular arrangement, where vertices connected by edges means two students are adjacent. A teacher has prepared multiple copies of 4 sets of questions. He wants to ensure that no two students sitting beside each other will get the same set of question. But it is completely fine to reuse copies of the same set of questions for students who are non-adjacent. The teacher wants to find out the number of different ways of assigning questions to the students.

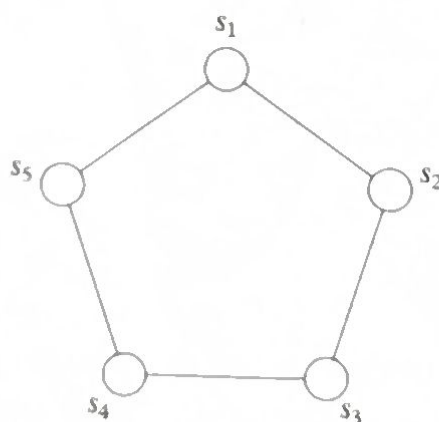


Figure 2: Sample graph representing a seating arrangement of five students for Question 4

i. Formulate a function which will output the total number of ways of assigning the questions, given the number of unique sets of questions.

ii. Using the formulation in Question 4.a)i, find the total number of ways of assigning the questions given 4 unique sets. If the teacher had prepared 3 sets instead of 4, would it still be possible to assign the questions such that no two adjacent students get the same question? 3(2, 26)

5 + 3
(CO3)
(PO3)

b) Justify the statement: "In all simple graphs of two or more vertices, there will always be at least 2 vertices having same degree."

5
(CO1)
(PO1)

5. Figure 3 shows a flow network where the weights associated with each edge represents the corresponding capacity of that edge.

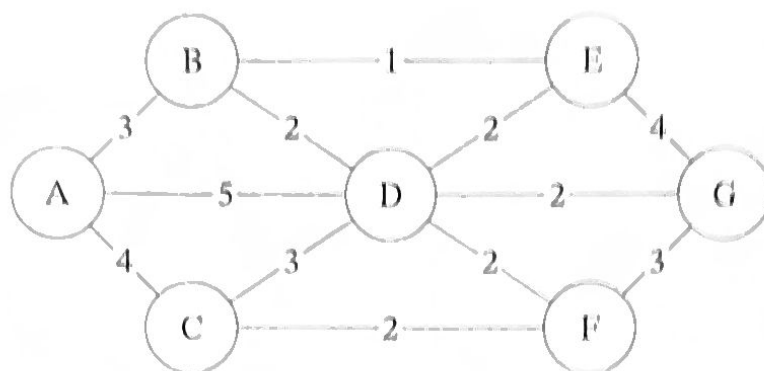


Figure 3: Sample graph representing a flow network for Question 5

a) Starting from any valid spanning tree of the flow network, generate all the fundamental cutsets with respect to that spanning tree.

10
(CO1)
(PO1)

b) Illustrate a method for generating all possible cutsets of the flow network using the fundamental cutsets already generated in Question 5.a.

10
(CO1)
(PO1)

c) Consider A as the source, and G as the sink. Filter the cutsets generated in Question 5.b and apply the Max-flow Min-cut theorem to find the maximum flow for the flow network.

5
(CO1)
(PO1)

6. a) Derive the expression for finding the nullity of a complete undirected graph of n vertices.

10
(CO1)
(PO1)

$$M = \frac{(n-1)(n-2)}{2}$$

b) Figure 4 shows a sample labelled graph G . Construct the tree graph, $T(G)$, for the given graph, where the vertices will represent the distinct spanning trees of G , and the edges will represent the cyclic interchange operation required to go from one spanning tree of G to another.

15
(CO1)
(PO1)

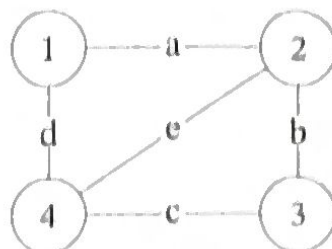


Figure 4: Sample graph for Question 6.b